The Cloud In The Bottle

Overview
Students are already interested in the weather and can be made increasingly aware of cloud formation by this rather simple in-class demonstration.

Purpose
The purpose to this activity is to demonstrate to the students the direct effects of pressure and temperature on cloud formation.

Objectives
1. Students will view an illustration of the principles involved in cloud formation.
2. Students will apply these principles to weather maps to predict cloud formation and clearing.
3. Students will observe how nuclei are necessary for the attachment of condensation in order to form clouds.

Resources/Materials
1. Wide mouth gallon pickle jar
2. Heavy duty clear plastic bag
3. Rubber bands or masking tape
4. Matches

Activities and Procedures
1. Place about 20 ml of water in a wide mouth gallon pickle jar.
2. Place a lit match into the jar.
3. Quickly place a heavy duty clear plastic bag over the mouth of the jar and secure a firm seal by placing a rubber band/masking tape around the top of the jar.
4. Push the bag into the jar quickly, then pull the bag out. Observe!

Tying it all Together
This activity illustrates how humidity, temperature, and air pressure influence the formation of clouds. The water produces high humidity in the jar and the smoke introduced by the match provides nuclei on which the water vapor can condense. As the bag is pushed into the jar, the pressure and temperature in the jar increase, causing the jar to clear. Upon pulling the bag out, pressure and temperature decrease, allowing water vapor to condense and produce a "cloud" inside the jar.

Questions For Further Inquiry
1. Why are smoke and water added to the jar? (They provide moisture and nuclei for cloud formation.)
2. What does pushing and pulling on the bag do to the jar? (They increase/decrease pressure.)
3. How would warming or cooling the jar affect the observations? (They decrease/increase cloud formation.)
4. Relate your findings to weather conditions necessary to produce cloudy/clear skies.

For articles, maps, or other media pertaining to this topic, look up the following in Encarta® Encyclopedia:

Cloud
Meteorology
Troposphere

For additional online information and media, view the following Encarta articles:
Cloud
Meteorology
Troposphere

For additional relevant information, visit the following Web sites:
Cloud Catalog
http://covis.atmos.uiuc.edu/guide/clouds/html/cloud.home.html